JINHAN KIM

AI and Software Engineering

Curriculum Vitae / NOV 2026

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01 APPOINTMENTS

Postdoctoral Researcher **Sep 2023 – Present**

Università della Svizzera italiana (USI), Lugano, Switzerland / Advisor: Prof. Paolo Tonella / Funding: the European Horizon project Sec4AI4Sec and the NRF Korea Postdoctoral Fellowship Program 2024 / Research on testing and assurance for deep-learning and RAG systems.

Postdoctoral Researcher **Mar 2023 – Aug 2023**

KAIST, Daejeon, South Korea / Advisor: Prof. Shin Yoo / Postdoctoral research at the COINSE lab.

Visiting Ph.D. Student **Aug 2022 – Sep 2022**

Università della Svizzera italiana (USI), Lugano, Switzerland / Advisor: Prof. Paolo Tonella / Funding: Young Researchers' Exchange Programme between South Korea and Switzerland 2022 / Research on mutation-based deep-learning system testing and deep-learning program repair.

Research Intern, Users & Information Lab **Sep 2015 – Dec 2015**

KAIST, Daejeon, South Korea / Advisor: Prof. Alice Oh / Designed and deployed EliceQ, a social back-channel application for anonymous classroom questions.

Research Intern, NC Lab **Mar 2015 – Aug 2015**

KAIST, Daejeon, South Korea / Advisor: Prof. Junehwa Song / Developed BeUpright and supported a two-week human study on relational-norm interventions for posture change.

02 EDUCATION

Ph.D. in Computer Science **Mar 2017 – Feb 2023**

KAIST, Daejeon, South Korea

Integrated M.S./Ph.D. programme. Advisor: Prof. Shin Yoo.

B.S. in Computer Science **Mar 2012 – Feb 2017**

KAIST, Daejeon, South Korea

03 PH.D. SUPERVISION

Masoud Jamshidiyan Tehrani **Feb 2024 – Present**

Ph.D. student at USI; co-supervised with Prof. Paolo Tonella

Samuele Pasini **Oct 2023 – Present**

Ph.D. student at USI; co-supervised with Prof. Paolo Tonella

04 AWARDS, HONORS & PATENT

2023 **Best Paper Award** 18th International Workshop on Mutation Analysis

2023 **KAIST Ph.D. Dissertation Award** College of Engineering, KAIST

2020 **NAVER Ph.D. Fellowship Award** School of Computing, KAIST

2021 **Korea Patent No. 1020190104591** *Method for Evaluating Test Fitness of Input Data for Neural Network and Apparatus Thereof*; published August 9, 2021.

05 PUBLICATIONS

Selected papers appear first; Jinhan Kim appears in navy. All are main-track unless noted.

Google Scholar | 12 Aug 2026 | Citations **1,067** | h-index **12** | i10-index **14**

SELECTED PUBLICATIONS

[1] ICSE 2027 **Rethinking Mutation Testing When the Natural Language Specification Is the Program**
J. Kim and P. Tonella
Proceedings of the 49th IEEE/ACM International Conference on Software Engineering.

[2] EMSE 2026 **Real Faults in Deep Learning Fault Benchmarks: How Real Are They?**
G. Jahangirova, N. Humbatova, **J. Kim**, S. Yoo, and P. Tonella
Empirical Software Engineering.

[3] ASE 2026 **When Knowledge Changes: Metamorphic Testing of RAG Systems with Mutations**
J. Kim, S. Pasini, and P. Tonella
Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering.

[4] ISSTA 2026 **Testing Retrieval-Augmented Generation Systems with Chunk Coverage**
J. Kim, S. Pasini, and P. Tonella
Proceedings of the 35th ACM SIGSOFT International Symposium on Software Testing and Analysis.

[5] ICSE 2026 **Revisiting "Revisiting Neuron Coverage for DNN Testing: A Layer-Wise and Distribution-Aware Criterion": A Critical Review and Implications on DNN Coverage Testing**
J. Kim, N. Humbatova, G. Jahangirova, S. Yoo, and P. Tonella
Proceedings of the 48th IEEE/ACM International Conference on Software Engineering.

[6] TOSEM 2025 **A Taxonomy of System-Level Attacks on Deep Learning Models in Autonomous Vehicles**
M. J. Tehrani, **J. Kim**, R. Z. L. Foulefack, A. Marchetto, and P. Tonella
ACM Transactions on Software Engineering and Methodology.
∗ *Journal-first presentation at ICSE 2026*

[7] EMSE 2025 **Evaluating and Improving the Robustness of Security Attack Detectors Generated by LLMs**
S. Pasini, **J. Kim**, T. Aiello, R. Lozoya, A. Sabetta, and P. Tonella
Empirical Software Engineering.
∗ *Journal-first presentation at ICSE 2026*

[8] EMSE 2025 **An Empirical Study of Fault Localisation Techniques for Deep Neural Networks**
N. Humbatova, **J. Kim**, G. Jahangirova, S. Yoo, and P. Tonella
Empirical Software Engineering, 30(5), p. 124.
∗ *Journal-first presentation at ICST 2026*

[9] TOSEM 2023 **Evaluating Surprise Adequacy for Deep Learning System Testing**
J. Kim, R. Feldt, and S. Yoo
ACM Transactions on Software Engineering and Methodology, 32(2), pp. 1–29.

[10] ICST 2023 **Repairing DNN Architecture: Are We There Yet?**
J. Kim, N. Humbatova, G. Jahangirova, P. Tonella, and S. Yoo
Proceedings of the 16th IEEE International Conference on Software Testing, Verification and Validation, pp. 234–245.

[11] TOSEM 2022 **Predictive Mutation Analysis via Natural Language Channel in Source Code**
J. Kim, J. Jeon, S. Hong, and S. Yoo
ACM Transactions on Software Engineering and Methodology, 31(4), pp. 1–27.
∗ *Journal-first presentation at ICSE 2023*

[12] ICSE 2019 **Guiding Deep Learning System Testing Using Surprise Adequacy**
J. Kim, R. Feldt, and S. Yoo
Proceedings of the 41st IEEE/ACM International Conference on Software Engineering, pp. 1039–1049.

OTHER JOURNAL ARTICLES

[1] **JSS 2026** **Cross-Site Scripting Adversarial Attacks Based on Deep Reinforcement Learning: Evaluation and Extension Study**
S. Pasini, G. Maragliano, **J. Kim**, and P. Tonella
Journal of Systems and Software.

[2] **IST 2023** **Learning Test-Mutant Relationship for Accurate Fault Localisation**
J. Kim, G. An, R. Feldt, and S. Yoo
Information and Software Technology, 162, p. 107272.

OTHER CONFERENCES & WORKSHOPS

[1] **TRUST 2026** **Dynamic Deception: When Pedestrians Team Up to Fool Autonomous Cars**
M. Tehrani, M. Gabriel, **J. Kim**, and P. Tonella
Proceedings of the 1st International Workshop on Trustworthy and Responsible aUtonomous SysTems.

[2] **FSE Demo 2025** **PCLA: A Framework for Testing Autonomous Agents in the CARLA Simulator**
M. Tehrani, **J. Kim**, and P. Tonella
Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering, pp. 1040–1044.

[3] **Mutation 2023** **The Inversive Relationship Between Bugs and Patches: An Empirical Study**
J. Kim, J. Park, and S. Yoo
Proceedings of the 18th International Workshop on Mutation Analysis, pp. 314–323.

[4] **ICST Industry 2022** **Repairing Fragile GUI Test Cases Using Word and Layout Embedding**
J. Yoon, S. Chung, K. Shin, **J. Kim**, S. Hong, and S. Yoo
Proceedings of the 15th IEEE International Conference on Software Testing, Verification and Validation.

[5] **ISSRE 2021** **Ahead of Time Mutation Based Fault Localisation Using Statistical Inference**
J. Kim, G. An, R. Feldt, and S. Yoo
Proceedings of the 32nd International Symposium on Software Reliability Engineering, pp. 253–263.

[6] **FSE Industry 2020** **Reducing DNN Labelling Cost Using Surprise Adequacy: An Industrial Case Study for Autonomous Driving**
J. Kim, J. Ju, R. Feldt, and S. Yoo
Proceedings of the 28th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering, pp. 1466–1476.

[7] **GI 2018** **Comparing Line and AST Granularity Level for Program Repair Using PyGGI**
G. An, **J. Kim**, and S. Yoo
Proceedings of the 4th Genetic Improvement Workshop, pp. 19–26.

[8] **L@S 2018** **Elicast: Embedding Interactive Exercises in Instructional Programming Screencasts**
J. Park, Y. H. Park, **J. Kim**, J. Cha, S. Kim, and A. Oh
Proceedings of the Fifth Annual ACM Conference on Learning at Scale, pp. 1–10.

[9] **SSBSE 2018** **Learning Without Peeking: Secure Multi-Party Computation Genetic Programming**
J. Kim, M. G. Epitropakis, and S. Yoo
Proceedings of the 10th International Symposium on Search Based Software Engineering, pp. 246–261.

[10] **KCSE 2017** **PyGGI: Python General framework for Genetic Improvement**
G. An, **J. Kim**, S. Lee, and S. Yoo
Proceedings of Korea Software Congress, pp. 536–538.

[11] **SSBSE Short Papers 2017** **GPGPGPU: Evaluation of Parallelisation of Genetic Programming Using GPGPU**
J. Kim, J. Kim, and S. Yoo
Proceedings of the 9th International Symposium on Search Based Software Engineering, pp. 137–142.

[12] **CHI 2016** **BeUpright: Posture Correction Using Relational Norm Intervention**
J. Shin, B. Kang, T. Park, J. Huh, **J. Kim**, and J. Song
Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems, pp. 6040–6052.

[13] **SenSys Demo 2015** **Demo: Posture Correction Using Smartphone-Based Relational Intervention Model**
J. Shin, B. Kang, **J. Kim**, J. Huh, J. Song, and T. Park
Proceedings of the 13th ACM Conference on Embedded Networked Sensor Systems, pp. 495–496.

PREPRINTS & MANUSCRIPTS

[1] **arXiv 2026** **Detecting Trojaned DNNs via Spectral Regression Analysis**
S. Pasini, **J. Kim**, and P. Tonella
arXiv preprint.

[2] **arXiv 2025** **Fault Localisation and Repair for DL Systems: An Empirical Study with LLMs**
J. Kim, N. Humbatova, G. Jahangirova, S. Yoo, and P. Tonella
arXiv preprint.
∗ *Under revision at ACM Transactions on Software Engineering and Methodology*

[3] **arXiv 2025** **MuFF: Stable and Sensitive Post-training Mutation Testing for Deep Learning**
J. Kim, N. Humbatova, G. Jahangirova, S. Yoo, and P. Tonella
arXiv preprint.

[4] **arXiv 2025** **Mutation-Guided Test Generation for DNNs Using Diffusion**
S. Kim, **J. Kim**, V. Riccio, S. Yoo, and P. Tonella
arXiv preprint.

06 ACADEMIC SERVICE

Organising Committees

2027 ICST, Posters Track Co-chair
2026 ICST, Testing Tools and Data Showcase Track Co-chair; SSBSE, Publicity Co-chair; DeepTest, Workshop Co-organiser; SBFT, Workshop Co-organiser
2025 DeepTest, Workshop Co-organiser

Program Committees

2027 ICSE, Research Track; ASTA
2026 ASE, Research Track; ESEM, Research Track; ICST, Education Track; Mutation; RASE
2025 ISSTA, Research Track; ASE, NIER Track and Student Research Competition; SSBSE, Challenge Track; SBFT; Mutation; DeMeSSAI
2024 SCAM, Research Track; ICSE, Demonstrations Track; ASE, Demonstrations and NIER Tracks; DeepTest; Mutation
2023 Mutation; ASE, NIER Track; ICSME, Artifact Evaluation Track and ROSE Festival
2022 Mutation; ICST, Poster Track; ICSME, Registered Reports Track, Artifact Evaluation Track, and ROSE Festival
2021 Mutation; ICSME, Artifact Evaluation Track
2020 Mutation

Journal Boards & Reviewing

2026 Reviewer: Empirical Software Engineering
2025 TOSEM Board of Distinguished Reviewers; reviewer for TSE, TOSEM, EMSE, STVR, and IEEE Computer
2024 TOSEM Board of Distinguished Reviewers; reviewer for TSE, TOSEM, EMSE, JSEP, and JSS
2023 Reviewer for TSE, TOSEM, EMSE, and JSEP
2022 Reviewer for TOSEM and JSEP
2021 Reviewer for TOSEM and STVR
2020 Reviewer for IST and JSS

07 INVITED TALKS

2026 **Mutation Workshop, ICST 2026, Daejeon, South Korea** / Panellist.

2026 **Software Institute Seminar, USI, Lugano, Switzerland** / *Testing RAG Systems: Coverage-Guided Adequacy and Metamorphic Oracles*.

2026 **CHOOSE Forum, Bern, Switzerland** / *Breaking DNNs: Mutation Testing for Deep Neural Networks*.

2024 **Software Institute Seminar, USI, Lugano, Switzerland** / *When Simple is Better than Complex: Coverage and Mutation for DL Testing*.

2023 **Korea Conference on Software Engineering, Pyeongchang, South Korea** / Invited paper presentation: *Predictive Mutation Analysis via Natural Language Channel in Source Code*.

2022 **Korea Conference on Software Engineering, Pyeongchang, South Korea** / Invited paper presentation: *Ahead of Time Mutation Based Fault Localisation using Statistical Inference*.

2019 **Korea Software Congress, Pyeongchang, South Korea** / Invited paper presentation: *Guiding Deep Learning System Testing Using Surprise Adequacy*.

08 TEACHING

KAIST CS101: Introduction to Programming – Teaching Assistant, Fall 2016, Spring 2020, Fall 2020, Spring 2021.
KAIST CS453: Automated Software Testing – Teaching Assistant, Spring 2018, Spring 2019.
KAIST CS454: AI Based Software Engineering – Teaching Assistant, Fall 2021.
KAIST CS489: Computer Ethics and Social Issues – Teaching Assistant, Fall 2019.

09 REFERENCES

Prof. Shin Yoo	Full Professor, KAIST, Daejeon, South Korea	shin.yoo@kaist.ac.kr
Prof. Paolo Tonella	Full Professor, USI, Lugano, Switzerland	paolo.tonella@usi.ch
Prof. Robert Feldt	Full Professor, Chalmers, Gothenburg, Sweden	robert.feldt@chalmers.se